

DiTOX: Fault Detection and Localization in the ONNX Optimizer



Nikolaos Louloudakis

n.louloudakis@ed.ac.uk

University of Edinburgh



Ajitha Rajan

arajan@ed.ac.uk

University of Edinburgh

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AI Compilers and Challenges in Testing Them

AI Compilers enable efficient deployment of AI models, bridging gap between ML frameworks and hardware while reducing latency, memory use and energy consumption.

Testing Challenges

- Oracle Problem: What is the '*correct*' output?
- Silent accuracy degradation after optimization
- Numerical instability and precision loss
- Large and complex search space of transformations

Compiler Specific Challenges

- Miscompilation of graph rewrites
- Backend-specific bugs
- Undefined behaviour in low-level kernels
- Memory layout and buffer management errors

The ONNX Standard

Open Standard: Aimed for portable machine learning models.

Strong Interoperability: Enables cross-framework use, PyTorch and TensorFlow included.

Graph-Based Format: Defines computation graphs with standardized operators and data types.

Strong Partners: Supported by Microsoft, Meta, NVIDIA, and Amazon.

Simplified Deployment: Simplifies running models across runtimes and hardware.

Effective Optimization: ONNX Optimizer provides reusable graph optimization passes.

Wide Popularity: ONNX Optimizer has approximately 800 GitHub stars.

Optimizations in DNNs

- **Graph-Based Optimizations:** Transform computation graphs (fusion, pruning, constant folding)
- **Computation-Based Optimizations:** Improve math execution (quantization, mixed precision, kernel tuning)
- **Memory-Based Optimizations:** Reduce memory usage and data movement (buffer reuse, layout optimization)
- **Hardware-Aware Optimizations:** Tailor execution to accelerators (GPU/TPU kernels, operator mapping)
- **Algorithmic Optimizations:** Change model structure while preserving accuracy (distillation, low-rank approximations)

Challenges in Graph-Based Optimizations

- **Scope:** Huge variety of graph structures possible
- **Equivalence:** Must preserve semantics, not structure visually
- **Dynamics:** Shapes and control flow vary widely
- **Interactions:** Optimization passes can conflict unexpectedly
- **Backend:** Operator behaviors differ across frameworks
- **Oracle:** No clear reference for correctness testing

To effectively mitigate these challenges and test the ONNX Optimizer, we propose **DiTOX**.

Related Work



Existing work on compiler optimization testing relies on compiler fuzzing and differential testing using randomly generated program



Existing tools primarily rely on crash bugs rather than label accuracy related bugs

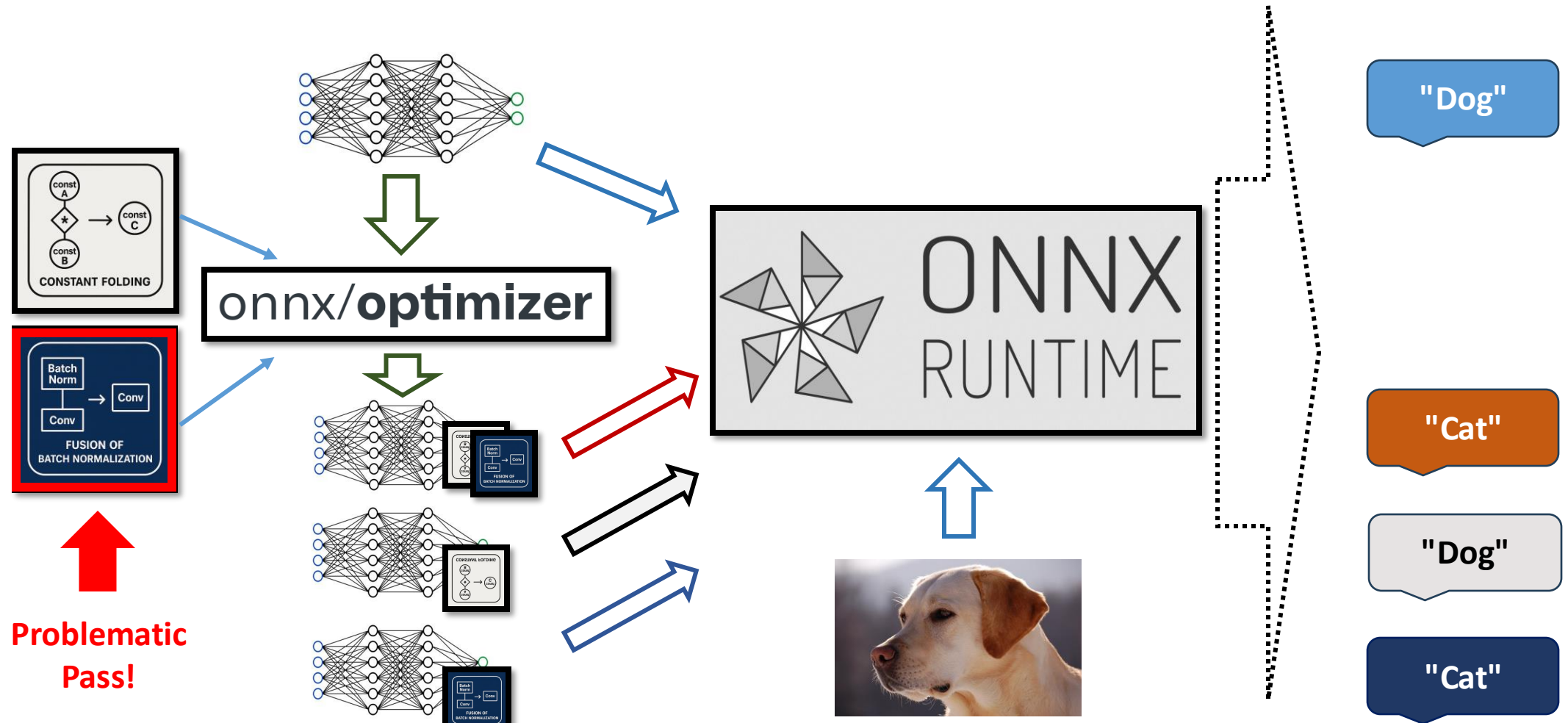


Support for graph optimization passes is limited

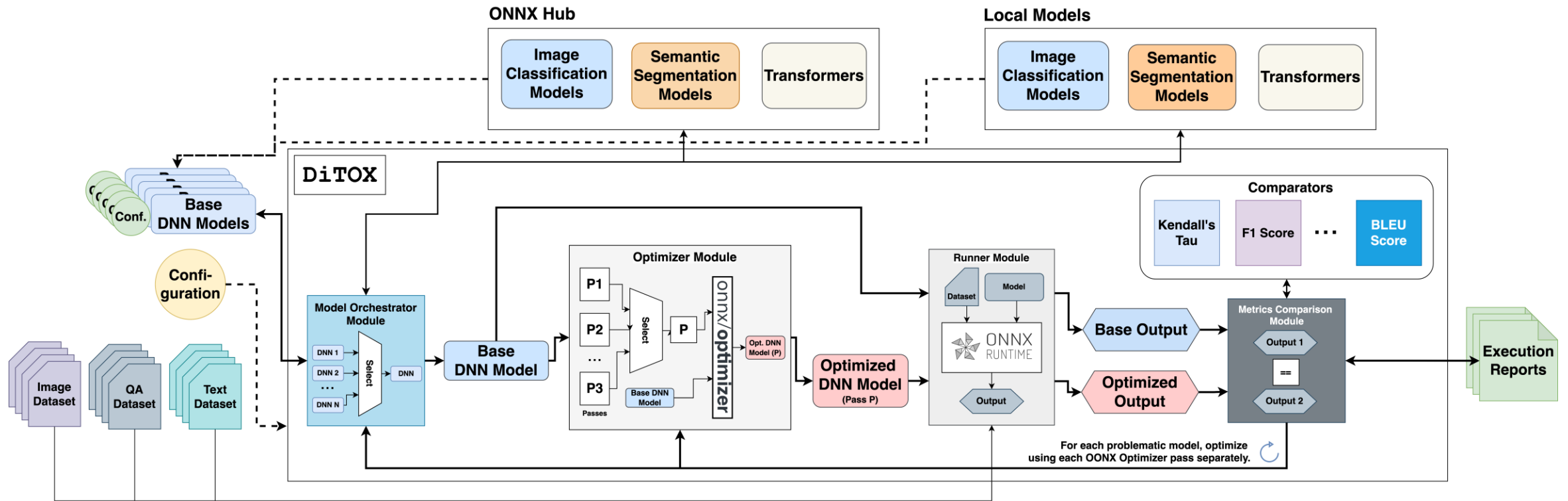


Limited support for ONNX optimization testing

DiTOX: The Differential Testing Process



DiTOX: Differential Testing of the ONNX Optimizer



Purpose: Detect and localize issues related to model optimization, introduced by the ONNX optimizer.

- Utilize automatic model fetching from official repositories.
- Optimize the models.
- Execute against the appropriate dataset.
- Perform differential testing using the suitable comparator based on the model type.
- Perform per-pass fault localization.

DiTOX: Features

Versatile Optimization:

- Enables full & partial model optimization.
- Utilizes both settings interchangeably to localize problematic passes

Strong Model Support:

- Automatic model fetching from official, external repositories (e.g., ONNX Model Hub) and local instances.
- Supports different types of ONNX models (e.g., Image Classifiers, Object Detectors, Transformers).

Different Issue Type Support:

- Identifies both crash and accuracy issues.

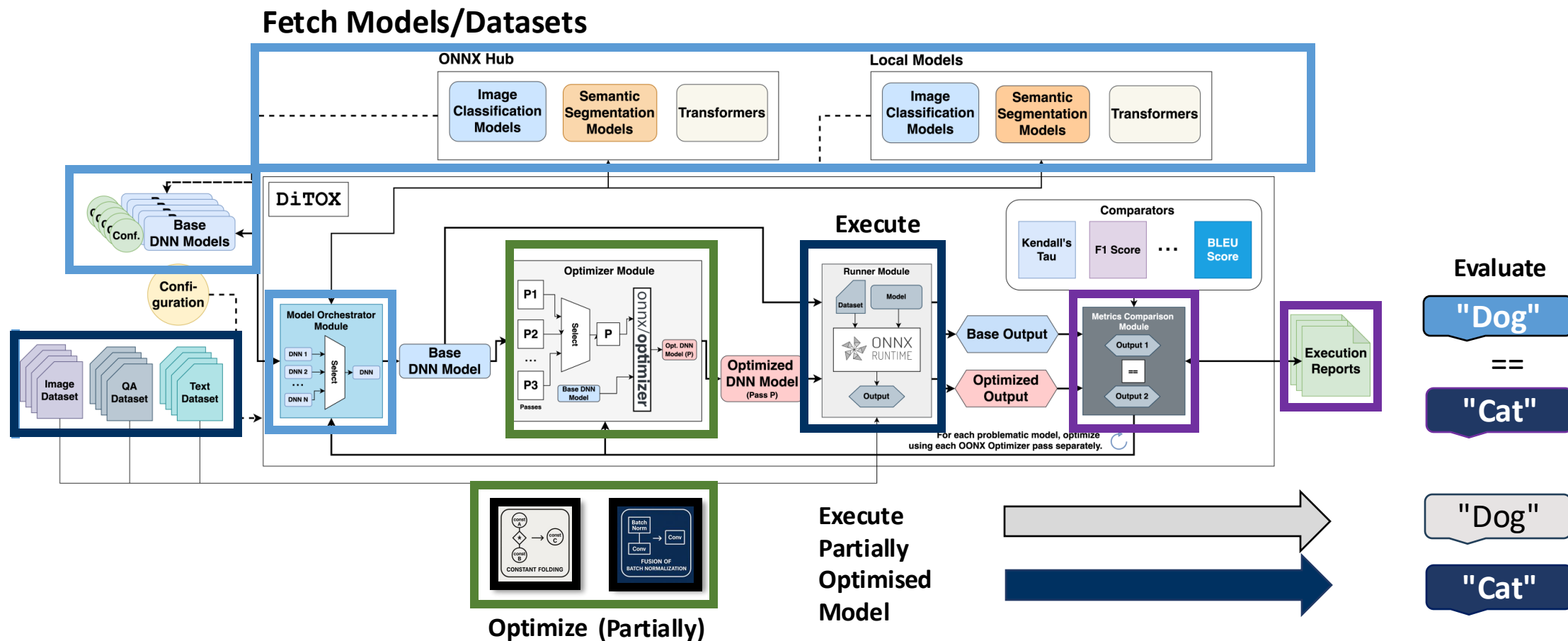
Appropriate Comparator Usage:

- Utilizes the appropriate comparator based on model type (e.g., Kendall's Tau for image classifiers, BLEU Score for text generators).

Effective Configuration:

- Configurable and extensible for new model types.

DiTOX: System Workflow



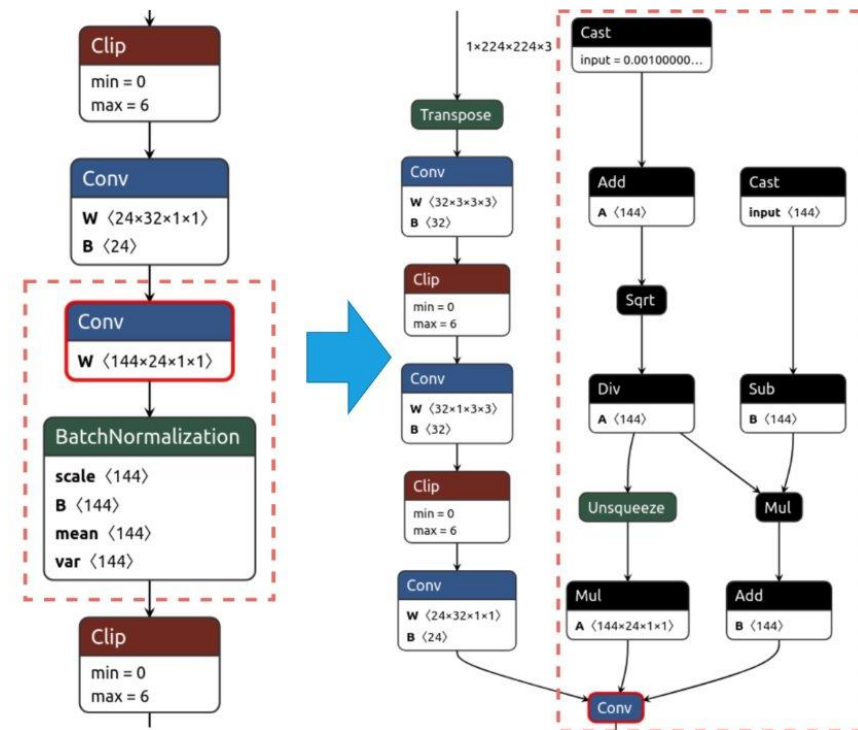
Experiments

Examined 130 Model Instances

- Obtained from the official ONNX Model Repository
 - 93 Classifiers** (e.g., MobileNet)
 - 30 Object Detectors** (e.g., YOLOv3)
 - 7 Transformers/Text Generators** (e.g., RoBERTa, GPT-2)Utilized models of a wide variety of opsets (7-12).
- Datasets:
 - ILSVRC2017** for image recognition for classification & object detection models,
 - Squad-V2** and **IMDB** for text generation models.
- Comparator Methods:
 - **Kendall's Tau** for classification
 - **F1 Score** (utilizing **IoU**), **Precision/Recall** for object detection
 - **BLEU score** for generic-purpose text generation.
 - **Tensor Comparison** for sentiment analysis.

DiTOX: Results (1/5)

- **Detected 15 Issues & Bugs**
 - 12 of them concerned crashes, 2 accuracy issues and 1 optimization warnings.
 - **14** of them were not in the issue tracker of the ONNX optimizer.
 - Provided unknown reproducibility details for the one previously reported issue.
 - Bugs detected were associated with:
 - 9 distinct passes
 - The way the optimizer internally handles ONNX format versions.



Detected graph issue in ***fuse_bn_into_conv*** pass.

DiTOX: Results (2/5)

#	Pass	Error	Type	Models
1	lift_lexical_references	Removes used value references.	Crash	EfficientNet-Lite4 (op. 11), SSD (op. 12)
2	split_init	Out of range List index	Crash	YOLOv3 (op. 10, 12)
3	split_init split_predict	Missing required field	Crash	DenseNet121 (op. 7-9), YOLOv2 (op. 10), SSD (op. 12), Tiny-YOLOv3 (op. 11), YOLOv3 (op. 10, 12)
4	split_predict	Missing required inputs	Crash	SSD (op. 12), DenseNet121 (op. 7), InceptionV2 (op. 6-9)
5	split_predict	Bad spec for node.	Crash	YOLOv3 (op. 10), DenseNet121 (op. 7), InceptionV2 (op. 6-9)
6	split_predict	Mismatch in #input elements	Crash	FCN ResNet-50-int8 (op. 12)
7	eliminate_shape_gather	Indices value $\neq$ node dims size.	Crash	Faster R-CNN: R-50-FPN (op. 10)
8	All Passes	Implicit ONNX format version update.	Acc.	Models of cases 1-11 in Table 3
9	rewrite_input_dtype	Multiple usage of output name	Crash	GPT-2 (op. 10)
10	eliminate_consecutive _idempotent_ops	Incompatible shape error.	Crash (Run)	Mask R-CNN/Faster R-CNN: R-50-FPN-int8 (12), R-50-FPN-qdq (op. 12), R-50-FPN-fp32 (op. 12)
11	eliminate_duplicate_initializer	Erase output assertion failed.	Crash	Faster R-CNN: R-50-FPN-qdq (12)
12	fuse_bn_into_conv	Duplicate initializer introduced.	Crash (Run)	RetinaNet (op. 9)
13	eliminate_shape_gather eliminate_nop_reshape	Unused initializers	Warn.	Faster R-CNN: R-50-FPN-int8 (op. 12)
14	All Passes	Invalid input scale.	Crash	VGG-16-BN (op. 7), VGG-19-BN (op. 7), ResNet101_DUC_HDC (op. 7)
15	fuse_bn_into_conv	Bad graph structure.	Acc.	Efficient-Lite4 (op. 11), SSD (op. 12)

List of Detected Issues

DiTOX: Results (3/5)

Classification Models:

- We tested 93 Classification model instances.
- **28** (30%) are found to have differences in top-10 (0.01 to 12.9%) label ranking.
- **12** (13%) are found to have differences in top-5 (0.01 to 1.92%) label ranking.
- None is found to have differences in top-1 label ranking.

#	Model	Type	Ver.	T-1	T-5	T-10
1	AlexNet [27]	CL	7-9	0%	0%	0.02%
2	DenseNet121 [24]	CL	7-9	0%	0%	0.02%
3	GoogleNet [61]	CL	7-9	0%	0%	0.04%
4	InceptionV1 [60]	CL	7-9	0%	0%	0.25%
5	InceptionV2 [60]	CL	7-9	0%	0.02%	0.05%
6	MobileNetV2 [23, 56]	CL	7-9	0%	0%	0.02%
7	ResNet18 [21]	CL	7	0%	0.02%	0.02%
8	ResNet34V2 [22]	CL	7	0%	0.02%	0.02%
9	ResNet50-caffe2 [21]	CL	7-9	0%	2%	13%
10	ShuffleNetV1 [70]	CL	7-9	0%	0.7%	1%
11	SqueezeNet 1.0 [25]	CL	7-9	0%	0%	0.02%
12	EfficientNet-Lite4 [1]	CL	11	0%	0.04%	1%
13	SSD [37]	OD	12	0%	0%	0.04%

DiTOX: Results (4/5)

Object Detection Models:

- We tested **30** model instances.
- Of those, **5** (16.6%) presented differences between the original and optimized.
- Up to **15.29%** differences in IoU, **0.6%** differences in mAP, **0.55%** in F1 score, precision and recall.

Model	Op.	F1	Prec.	Rec.	mAP	IoU
SSD [37]	12	99.99%	99.99%	99.99%	99.99%	100%
Tiny YOLOv3 [51]	11	99.18%	99.18%	99.18%	99.99%	89.35%
YOLOv2 [50]	9	99.95%	99.95%	99.95%	99.94%	88.49%
YOLOv3 [51]	10	99.45%	99.45%	99.45%	99.99%	84.81%
YOLOv3 [51]	12	99.45%	99.45%	99.45%	99.99%	84.81%

DiTOX: Results (5/5)

Transformers/Text Generation Models:

- Tested **7** model instances, we detected **no significant accuracy differences**.
- We detected one crash issue in the ONNX optimizer associated with GPT-2 model.
- Models such as **T5**, **BERT-Squad** and **RoBERTa** presented no differences.
- While testing **GPT-2**, we detected only **8** samples out of **11873** of Squad-V2 that presented BLEU score differences.
- We considered the **top-3** token predictions for **GPT-2**.

Key Observations

Classification Models: Erroneous optimizations can affect label accuracy.

Legacy Versions: Implicitly and incorrectly applied by optimizer can affect model accuracy.

Object Detection & Segmentation: Higher crash likelihood due to more complex model structure.

Bounding Box Accuracy (Object Detectors): Occasional misalignments reduce IoU up to 15.19%.

Top-10 Accuracy (Object Detectors): Largely stable despite bounding box degradation.

Text Comprehension/Generation: Rare crashes, minor BLEU-score differences.

Overall: Susceptibility to optimizer issues varies by model type.

Tool Effectiveness: Can isolate passes causing discrepancies across models

Future Work

- **Hardware acceleration:** Extend **DiTOX** on other AI compilers (e.g., Apache TVM).
- **Model type selection:** Explore broader models (diffusion, RL models, speech).
- **Optimization pass order:** Study effects of different pass sequences.

Availability

- You can access DiTOX at <https://github.com/luludak/DiTOX>
- We have reported all issues at the official ONNX Optimizer Repository.
- Artifact is Available, Reusable, and Reproduced.

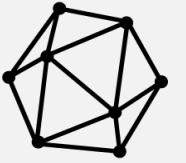


🟢	[BUG] Pass "fuse_bn_into_conv" results in problematic graph and affects results. #178 - nick-louloudakis opened on Mar 26, 2025
🟢	[BUG] Issue with BatchNormalization node dimensions on VGG and DUC_HDC models (opset=7) #177 - nick-louloudakis opened on Mar 26, 2025
🟢	Redundant initializers introduced to the models using eliminate passes. #176 - nick-louloudakis opened on Mar 26, 2025
🟢	Pass "fuse_bn_into_conv" introduces duplicate initializer. #174 - nick-louloudakis opened on Mar 26, 2025
🟢	Pass "eliminate_consecutive_idempotent_ops" succeeds but model gives an incompatible shape error upon run. #171 - nick-louloudakis opened on Mar 19, 2025
🟢	Pass "rewrite_input_dtype" gives an error on GPT-2. #170 - nick-louloudakis opened on Mar 18, 2025
🟢	[BUG] Pass "eliminate_shape_gather" results in a dimension size error. #168 - nick-louloudakis opened on Mar 6, 2025
🟢	The "split" passes result in invalid models. #167 - nick-louloudakis opened on Mar 4, 2025
🟢	The "lift_lexical_references" pass should not lead in an invalid model, but extract metadata instead. #166 - nick-louloudakis opened on Mar 3, 2025

Issues Reported at the ONNX Optimizer



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Thank You

